$$\begin{aligned} \max\quad & 4x_1+2x_2+4x_3 \\ \text{s.t.}\quad & 4x_1+x_2+2x_3 \le 24 \\ & 2x_1+5x_2+3x_3 \le 30 \\ & 2x_2+x_3 \le 16 \\ & x_1,\ x_2,\ x_3 \ge 0 \end{aligned}$$

Introduce slack variables $s_1, s_2, s_3 \ge 0$

$$\begin{aligned} \max\quad & 4x_1+2x_2+4x_3 \\ \text{s.t.}\quad & 4x_1+x_2+2x_3+s_1=24 \\ & 2x_1+5x_2+3x_3+s_2=30 \\ & 2x_2+x_3+s_3=16 \\ & x_1, x_2, x_3, s_1, s_2, s_3 \ge 0 \end{aligned}$$

Canonical form:

$$\begin{aligned} \max\quad & z=c^T x \\ \text{s.t.}\quad & Ax=b \\ & x \ge 0 \end{aligned}$$

Basis:

$$x=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} x_N \\ x_B \end{bmatrix}$$

$$A=\begin{bmatrix} 4 & 1 & 2 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 0 & 2 & 1 & 0 & 0 & 1 \end{bmatrix} = [N\ B]$$

$$N=\begin{bmatrix} 4 & 1 & 2 \\ 2 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix},\quad B=I$$

$$b=\begin{bmatrix} 24 \\ 30 \\ 16 \end{bmatrix}$$

$$c^T=[4\ 2\ 4\ 0\ 0\ 0]=[c_N^T\ \ c_B^T]$$

$$c_N^T=[4\ 2\ 4],\quad c_B^T=[0\ 0\ 0]$$

Current basic and nonbasic solutions:

$$x_B=\begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}=B^{-1}b=Ib=\begin{bmatrix} 24 \\ 30 \\ 16 \end{bmatrix},\quad x_N=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=0$$

Optimality check:

Simplex multipliers:

$$\pi^T=c_B^T B^{-1}=[0\ 0\ 0]\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}=[0\ 0\ 0]$$

Reduced cost:

$$\bar{c}_N^T=c_N^T-c_B^T B^{-1}N=c_N^T-\pi^T N$$

$$=[4\ 2\ 4]-[0\ 0\ 0]\begin{bmatrix} 4 & 1 & 2 \\ 2 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix}=[4\ 2\ 4]$$

Entering variable $x_e=x_1$

Simplex direction: $d_N=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Minimum ratio test:

compute the simplex direction:

$$d_B^{(e)}=-B^{-1}a_e=-\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix}=\begin{bmatrix} -4 \\ -2 \\ 0 \end{bmatrix}$$

$$x_B=\begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}=\begin{bmatrix} 24 \\ 30 \\ 16 \end{bmatrix}$$

Minimum ratio:

$$\lambda=\min\left\{\frac{x_j}{-d_j}: d_j<0,\ j\in B\right\}$$

$$=\min\left\{\frac{24}{4},\frac{30}{2}\right\}=6$$

Leaving variables $x_l=s_1$

Update solution and basis:

Update the solution: $x^{new}=x+\lambda d$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \\ 24 \\ 30 \\ 16 \end{bmatrix}+6\begin{bmatrix} 1 \\ 0 \\ 0 \\ -4 \\ -2 \\ 0 \end{bmatrix}=\begin{bmatrix} 6 \\ 0 \\ 0 \\ 0 \\ 18 \\ 16 \end{bmatrix}$$

New basic and nonbasic variables:

$$x_B=\begin{bmatrix} x_1 \\ s_2 \\ s_3 \end{bmatrix}=\begin{bmatrix} 6 \\ 18 \\ 16 \end{bmatrix},\quad x_N=\begin{bmatrix} x_2 \\ x_3 \\ s_1 \end{bmatrix}=0$$

Update the basis:

$$c_B^T=[4\ 0\ 0],\quad c_N^T=[0\ 2\ 4]$$

$$B=\begin{bmatrix} 4 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},\quad N=\begin{bmatrix} 1 & 1 & 2 \\ 0 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix}$$

$$b=\begin{bmatrix} 24 \\ 30 \\ 16 \end{bmatrix}$$

Basic and nonbasic variables:

$$B^{-1}=\begin{bmatrix} 4 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1}=\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x_B=\begin{bmatrix} x_1 \\ s_2 \\ s_3 \end{bmatrix}=B^{-1}b=\begin{bmatrix} 6 \\ 18 \\ 16 \end{bmatrix},\quad x_N=0$$

$$c_B^T=[4\ 0\ 0],\quad c_N^T=[0\ 2\ 4]$$

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Optimality check:

Simplex multipliers:

$$\pi^T = C_B^T B^{-1} = [4\ 0\ 0]\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [1\ 0\ 0]$$

Reduced cost:

$$\overline{C_N}^T = C_N^T - C_B^T B^{-1} N = C_N^T - \pi^T N$$

$$= [0\ 2\ 4] - [1\ 0\ 0]\begin{bmatrix} 1 & 1 & 2 \\ 0 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix}$$

$$= [0\ 2\ 4] - [1\ 1\ 2]$$

$$= [-1\ 1\ 2]$$

Entering variable $x_e = x_3$

Simplex direction: $d_N = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Minimum ratio test:

Compute the simplex direction:

$$d_B^{(e)} = -B^{-1} a_e = -\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -2 \\ -1 \end{bmatrix}$$

$$x_B = \begin{bmatrix} x_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 18 \\ 16 \end{bmatrix}$$

Minimum ratio:

$$\lambda = \min\left\{\frac{x_j}{-d_j} : d_j < 0,\ j \in B\right\} = \min\left\{\frac{6}{\frac{1}{2}}, \frac{18}{2}, \frac{16}{1}\right\}$$

$$= 9$$

Leaving variables $x_l = S_2$

Update solution and basis:

Update the solution: $x^{new} = x + \lambda d$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \\ 0 \\ 18 \\ 16 \end{bmatrix} + 9\begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ 0 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 0 \\ 9 \\ 0 \\ 0 \\ 7 \end{bmatrix}$$

New basic and nonbasic variables:

$$x_B = \begin{bmatrix} x_1 \\ x_3 \\ S_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 9 \\ 7 \end{bmatrix},\quad x_N = \begin{bmatrix} S_1 \\ x_2 \\ S_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Update the basis:

$$C_B^T = [4\ 4\ 0],\quad C_N^T = [0\ 2\ 0]$$

$$B = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 1 & 1 \end{bmatrix},\quad N = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 2 & 0 \end{bmatrix},\quad B^{-1} = \begin{bmatrix} \frac{3}{8} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & \frac{1}{2} & 0 \\ \frac{1}{4} & -\frac{1}{2} & 1 \end{bmatrix}$$

$$\cancel{x_B = \begin{bmatrix} x_1 \\ x_3 \\ S_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 9 \\ 7 \end{bmatrix}}\quad \cancel{x_N = \begin{bmatrix} S_1 \\ x_2 \\ S_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

$$\cancel{C_B^T = [4\ 4\ 0],\quad C_N^T = [0\ 2\ 0]}$$

Optimality check:

Simplex multipliers:

$$\pi^T = C_B^T B^{-1} = [4\ 4\ 0]\begin{bmatrix} \frac{3}{8} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & \frac{1}{2} & 0 \\ \frac{1}{4} & -\frac{1}{2} & 1 \end{bmatrix}$$

$$= [\tfrac{1}{2}\ \ 1\ \ 0]$$

Reduced cost:

$$\overline{C_N}^T = C_N^T - C_B^T B^{-1} N = C_N^T - \pi^T N$$

$$= [0\ 2\ 0] - [\tfrac{1}{2}\ 1\ 0]\begin{bmatrix} 1 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

$$= [0\ 2\ 0] - [\tfrac{1}{2}\ \tfrac{11}{2}\ 1]$$

$$= [-\tfrac{1}{2}\ -\tfrac{7}{2}\ -1]$$

No improving directions

Optimal solution:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 0 \\ 9 \\ 0 \\ 0 \\ 7 \end{bmatrix}$$

$$\begin{cases} x_1 = \frac{3}{2} \\ x_2 = 0 \\ x_3 = 9 \\ Z = 4 \times \frac{3}{2} + 2 \times 0 + 9 \times 4 = 42 \end{cases}$$

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